Étale Coverings and the Fundamental Group

(SGA 1)

English source-synchronization workpass

Exposé I opening through Proposition I.9.1

Task-owned working translation

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Source and status note

This bounded workpass translates the opening of Exposé I through Proposition I.9.1 directly from the French TEX source in arXiv `math/0206203v2`, with archive SHA-256:

`0A33C9C06908705A2525690FAAE02F6F`
`07980A4AA069A8B6CFF9B1D9BC39ACD3`

The current translated source boundary is `smf_doc-math_3_01.tex`, lines 556–1703; line 1704 begins Corollary I.9.2 and is outside the current workpass. The opening through § I.8 is preserved in the immutable, independently audited r4 checkpoint; its exact custody payload has been dispatched to the English/Germanic manager and archive maintenance. Parent custody acknowledgment, archive acceptance, publication, and remote readback are not claimed here. The § I.9 opening through Proposition I.9.1 is the current source-, formula-, structure-, terminology-, build-, and render-review unit.

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Contents

CHAPTER I

Étale morphisms

To simplify the exposition, we assume that all preschemes under consideration are locally noetherian, at least after no. I.2. French source p. 1

I.1. Notions of differential calculus

Let X be a prescheme over Y, and let $\Delta_{X/Y}$, or simply Δ, be the diagonal morphism $X \to X \times_Y X$. It is an immersion, hence a closed immersion of X into an open subset V of $X \times_Y X$. Let $\mathcal{I}_X$ be the ideal of the closed subprescheme corresponding to the diagonal in V (N.B. if one wishes to proceed intrinsically, without assuming that X is separated over Y—a hypothesis that would be farcical—one should consider the set-theoretic inverse image of $\mathcal{O}_{X\times X}$ in X, and denote by $\mathcal{I}_X$ the augmentation ideal in the latter...). The sheaf $\mathcal{I}_X/\mathcal{I}_X^2$ may be regarded as a quasi-coherent sheaf on X; we denote it by $\Omega^1_{X/Y}$. It is of finite type if $X \to Y$ is of finite type. It behaves well under base change $Y' \to Y$. We also introduce the sheaves $\mathcal{O}_{X\times_Y X}/\mathcal{I}_X^{n+1} = \mathcal{P}^n_{X/Y}$; these are sheaves of *rings* on X, making X into a prescheme that may be denoted by $\Delta^n_{X/Y}$ and called the *n-th infinitesimal neighborhood of* X/Y. The resulting sorites is entirely trivial, although rather long[1]; it would be prudent to speak of it only when there is something useful to say about it, in connection with smooth morphisms.

I.2. Quasi-finite morphisms

Proposition I.2.1. *Let $A \to B$ be a local homomorphism (N.B. the rings are now noetherian), and let $\mathfrak{m}$ be the maximal ideal of A. The following conditions are equivalent:*

(i) *$B/\mathfrak{m}B$ is finite-dimensional over $k = A/\mathfrak{m}$.*
(ii) *$\mathfrak{m}B$ is an ideal of definition, and $B/\mathfrak{r}(B) = \kappa(B)$ is an extension of $k = \kappa(A)$.*
(iii) *The completion $\widehat{B}$ is finite over the completion $\widehat{A}$ of A.*

One then says that B *is quasi-finite* over A. A morphism $f\colon X \to Y$ is said to be quasi-finite at x (or the Y-prescheme f is said to be quasi-finite at x) if $\mathcal{O}_x$ is quasi-finite over $\mathcal{O}_{f(x)}$. This is also equivalent to saying that x is *isolated in its fiber* $f^{-1}(f(x))$.[2] A morphism is said to be quasi-finite if it is so at every point.[3] French source p. 2

Corollary I.2.2. *If A is complete, quasi-finite is equivalent to finite.*

One could give the usual chain of implications (i), (ii), (iii), (iv), (v) for quasi-finite morphisms, but this does not seem indispensable here.

I.3. Unramified or net morphisms

Proposition I.3.1. *Let $f\colon X \to Y$ be a morphism of finite type, let $x \in X$, and put $y = f(x)$. The following conditions are equivalent:*

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[1]Cf. EGA IV 16.3.

[2]The original printing (Exposé I, p. 2) and both French arXiv renderings print $f^{-1}(x)$. Since $x \in X$ for $f\colon X \to Y$, the type-correct fiber through x is $f^{-1}(f(x))$, which is used here with the source defect explicitly disclosed.

[3]In EGA II 6.2.3 one assumes in addition that f is of finite type.

(i) *$\mathcal{O}_x/\mathfrak{m}_y\mathcal{O}_x$ is a finite separable extension of $\kappa(y)$.*
(ii) *$\Omega^1_{X/Y}$ vanishes at x.*
(iii) *The diagonal morphism $\Delta_{X/Y}$ is an open immersion in a neighborhood of x.*

For the implication (i)⇒(ii), Nakayama's lemma immediately reduces us to the case $Y = \operatorname{Spec}(k)$, $X = \operatorname{Spec}(k')$, where the assertion is well known and, moreover, immediate from the definition of separability. The implication (ii)⇒(iii) follows from a pleasant and easy characterization of open immersions using Krull. For (iii)⇒(i), one is again reduced to the case $Y = \operatorname{Spec}(k)$, with the diagonal morphism an open immersion everywhere. One must then prove that X is finite, with a separable coordinate ring over k; for this one reduces to the case where k is algebraically closed. But then every closed point of X is isolated (it is precisely the inverse image of the diagonal under the morphism $X \to X \times_k X$ defined by x), and hence X is finite. One may therefore suppose that X consists of a single point, with ring A. Thus $A \otimes_k A \to A$ is an isomorphism, whence $A = k$, as required.

Definition I.3.2. (a) *One then says that f is* net*, or equivalently* unramified*, at x, or that X is net, or unramified, at x over Y.*
(b) *Let $A \to B$ be a local homomorphism. It is said to be* net*, or* unramified*, and B is said to be a net, or unramified, local algebra over A, if $B/\mathfrak{r}(A)B$ is a finite separable extension of $A/\mathfrak{r}(A)$; that is, if $\mathfrak{r}(A)B = \mathfrak{r}(B)$ and $\kappa(B)$ is a separable extension of $\kappa(A)$.*[4]

Remarks. The fact that B is unramified over A can already be checked on the completions of A and B. Unramified implies quasi-finite. French source p. 3

Corollary I.3.3. *The set of points at which f is unramified is open.*

Corollary I.3.4. *Let X' and X be two preschemes of finite type over Y, and let $g\colon X' \to X$ be a Y-morphism. If X is unramified over Y, the graph morphism*

$$\Gamma_g\colon X' \longrightarrow X' \times_Y X$$

is an open immersion.[5]

Indeed, it is the inverse image of the diagonal morphism $X \to X \times_Y X$ under

$$g \times_Y \mathrm{id}_X\colon X' \times_Y X \longrightarrow X \times_Y X.$$

One may also introduce the annihilator ideal $\mathfrak{d}_{X/Y}$ of $\Omega^1_{X/Y}$, called the *different ideal* of X/Y. It defines a closed subprescheme of X whose underlying set is the set of points at which X/Y is ramified, that is, not unramified.

Proposition I.3.5. (i) *An immersion is unramified.*
(ii) *The composite of two unramified morphisms is unramified.*
(iii) *A base change of an unramified morphism is unramified.*

This may be seen equally well from criterion (ii) or criterion (iii) (the latter seems more amusing to me). One can of course also be more precise by giving pointwise statements; this is more general only in appearance (except in the setting of Definition I.3.2(b)), and tedious. As usual, one obtains the following corollaries.

Corollaries I.3.6. (iv) *The Cartesian product of two unramified morphisms is unramified.*
(v) *If gf is unramified, then f is unramified.*

[4]Cf. the regrets in Exposé III, no. 1.2.

[5]The original printing (Exposé I, p. 3) and the French TeX give $X \times_Y X$ as the graph target and later write $\mathrm{id}_{X'}$ in the pullback map. The typed fiber products force $X' \times_Y X$ and id_X, respectively; those type-correct readings are used here with both source defects disclosed.

(vi) *If f is unramified, then f_{red} is unramified.*

Proposition I.3.7. *Let $A \to B$ be a local homomorphism. Suppose that the residue-field extension $\kappa(B)/\kappa(A)$ is trivial, or that $\kappa(A)$ is algebraically closed. Then B/A is unramified if and only if $\widehat{B}$, as an $\widehat{A}$-algebra, is a quotient of $\widehat{A}$.*

Remarks.

- When the residue-field extension is not assumed to be trivial, one may reduce to that case by making a suitable finite flat extension over A that kills the extension in question.
- Take the example in which A is the local ring of an ordinary double point of a curve and B is the local ring at a point of the normalization. Then $A \subset B$, and B is unramified over A with trivial residue-field extension; the map $\widehat{A} \to \widehat{B}$ is surjective but *not injective*. We shall therefore strengthen the notion of unramifiedness. French source p. 4

I.4. Étale morphisms. Étale coverings

We shall take for granted whatever is needed concerning flat morphisms; these facts will be proved later, if necessary[6].

Definition I.4.1. (a) *Let $f\colon X \to Y$ be a morphism of finite type. One says that f is* étale *at x if f is flat at x and net (equivalently, unramified) at x. One says that f is étale if it is so at every point. One says that X is étale at x over Y, or that it is a Y-prescheme étale at x, and so on.*

(b) *Let $f\colon A \to B$ be a local homomorphism. One says that f is étale, or that B is étale over A, if B is flat and unramified over A.*[7]

Proposition I.4.2. *For B/A to be étale, it is necessary and sufficient that $\widehat{B}/\widehat{A}$ be étale.*

Indeed, this holds separately for "unramified" and for "flat."

Corollary I.4.3. *Let $f\colon X \to Y$ be of finite type, and let $x \in X$. Whether f is étale at x depends only on the local homomorphism $\mathcal{O}_{f(x)} \to \mathcal{O}_x$, and even only on the corresponding homomorphism between the completions.*

Corollary I.4.4. *Suppose that the residue-field extension $\kappa(A) \to \kappa(B)$ is trivial, or that $\kappa(A)$ is algebraically closed. Then B/A is étale if and only if $\widehat{A} \to \widehat{B}$ is an isomorphism.*

This follows by combining flatness with Proposition I.3.7.

Proposition I.4.5. *Let $f\colon X \to Y$ be a morphism of finite type. Then the set of points at which it is étale is open.*

Indeed, this holds separately for "unramified" and for "flat."

This proposition shows that one may dispense with "pointwise" statements when studying morphisms of finite type that are étale at some point.

Proposition I.4.6. (i) *An open immersion is étale.*

(ii) *The composite of two étale morphisms is étale.*

(iii) *A base change of an étale morphism is étale.*

French source p. 5

Indeed, (i) is trivial, and for (ii) and (iii) it is enough to note that the assertions hold for "unramified" and for "flat." To tell the truth, there are also corresponding statements for local homomorphisms (without a finiteness condition); in any event, they will have to appear in the *multiplodoque*, beginning with the unramified case.

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[6] Cf. Exposé IV.

[7] Cf. the regrets in Exposé III, no. 1.2.

COROLLARY I.4.7. *The Cartesian product of two étale morphisms is likewise étale.*

COROLLARY I.4.8. *Let X and X' be of finite type over Y, and let $g\colon X \to X'$ be a Y-morphism. If X' is unramified over Y and X is étale over Y, then g is étale.*

Indeed, g is the composite of the graph morphism

$$\Gamma_g\colon X \longrightarrow X \times_Y X',$$

which is an open immersion by Corollary I.3.4, and the projection $X \times_Y X' \to X'$, which is étale because it is obtained from the étale morphism $X \to Y$ by the base change $X' \to Y$.

DEFINITION I.4.9. *An* étale covering *(respectively, a* net covering*, also called an unramified covering) of Y is a Y-scheme X that is finite over Y and étale (respectively, unramified) over Y.*

The first condition means that X is defined by a coherent sheaf of algebras $\mathcal{B}$ on Y. The second then means that $\mathcal{B}$ is locally free over Y (with no corresponding local-freeness requirement in the net case), and, moreover, that for every $y \in Y$ the fiber

$$\mathcal{B}(y) = \mathcal{B}_y \otimes_{\mathcal{O}_y} \kappa(y)$$

is a separable algebra over $\kappa(y)$, that is, a finite product of finite separable field extensions of $\kappa(y)$.

PROPOSITION I.4.10. *Let X be a flat covering of Y of degree n (the definition of this term deserved to appear in Definition I.4.9), defined by a coherent locally free sheaf of algebras $\mathcal{B}$. In the usual way one defines the trace homomorphism $\mathcal{B} \to \mathcal{A}$, which is a homomorphism of $\mathcal{A}$-modules, where $\mathcal{A} = \mathcal{O}_Y$. For X to be étale, it is necessary and sufficient that the corresponding trace pairing $(x, y) \longmapsto \mathrm{Tr}_{\mathcal{B}/\mathcal{A}}(xy)$ induce an isomorphism $\mathcal{B} \xrightarrow{\sim} \mathcal{B}^\vee$,[8] or, equivalently, that the* discriminant section

$$d_{X/Y} = d_{\mathcal{B}/\mathcal{A}} \in \Gamma\Big(Y, \bigwedge^n \mathcal{B}^\vee \otimes_{\mathcal{A}} \bigwedge^n \mathcal{B}^\vee\Big)$$

be invertible, or, finally, that the discriminant ideal defined by this section be the unit ideal.

Indeed, one reduces to the case $Y = \mathrm{Spec}(k)$; the assertion is then the well-known separability criterion, and becomes immediate after passing to an algebraic closure of k. French source p. 6

Remark. There will be a less trivial statement below, when one does not assume a priori that X is flat over Y, but instead imposes a normality hypothesis.

I.5. The fundamental property of étale morphisms

THEOREM I.5.1. *Let $f\colon X \to Y$ be a morphism of finite type. For f to be an open immersion, it is necessary and sufficient that it be* étale and radicial.

Recall that radicial means injective, with radicial residue-field extensions; one may also recall that this means that the morphism remains injective after every base change. Necessity is trivial; it remains to prove sufficiency. We shall give two different proofs, the first shorter and the second more elementary.

1) A flat morphism is open, so (replacing Y by $f(X)$) we may suppose that f is a *homeomorphism* onto Y. After every base change, f remains flat, radicial, and surjective, hence a homeomorphism and therefore closed. Thus f is *proper*. Hence f is *finite* (reference: Chevalley's theorem) and is defined by a coherent sheaf $\mathcal{B}$ of algebras. The sheaf $\mathcal{B}$ is locally free and, by the hypothesis, has rank 1 everywhere. Thus $X = Y$, as required.

[8] The French TeX and the original printing instead give $\mathcal{B}$ as the target. An $\mathcal{A}$-bilinear form induces a map $\mathcal{B} \to \mathcal{B}^\vee$; the dual is therefore the type-correct target, used here with the source-original defect disclosed.

2) One may suppose Y and X *affine*. Moreover, one easily reduces to proving the following: if $Y = \mathrm{Spec}(A)$, with A local, and if $f^{-1}(y)$ is nonempty, where y is the closed point of Y, then $X = Y$. Indeed, this will imply that every $y \in f(X)$ has an open neighborhood U such that $X|U = U$. We shall have $X = \mathrm{Spec}(B)$, and we want to prove $A = B$. For this, it is enough to prove the analogous assertion after replacing A by $\widehat{A}$ and B by $B \otimes_A \widehat{A}$, taking into account that $\widehat{A}$ is faithfully flat over A. We may therefore suppose that A is *complete*. Let x be the point above y. By Corollary I.2.2, $\mathcal{O}_x$ is finite over A, and hence, being flat and radicial over A, is A itself. Thus

$$X = Y \amalg X'$$

as a disjoint union. Since X is radicial over Y, X' is empty. This completes the proof.

Corollary I.5.2. *Let $f\colon X \to Y$ be both a* closed immersion *and* étale. *If X is connected, f is an isomorphism from X onto a connected component of Y.*

Indeed, f is also an open immersion. It follows that: French source p. 7

Corollary I.5.3. *Let X be an unramified Y-scheme, with Y connected. Then every section of X over Y is an isomorphism from Y onto a connected component of X. There is therefore a one-to-one correspondence between the set of such sections and the set of connected components X_i of X for which the projection $X_i \to Y$ is an isomorphism (or, equivalently by Theorem I.5.1, is surjective, étale, and radicial).*[9] *In particular, a section is determined by its value at one point.*

Only the first assertion requires proof. By Corollary I.5.2, it is enough to observe that a section is a closed immersion (because X is separated over Y) and is étale by Corollary I.4.8.

Corollary I.5.4. *Let X and Y be preschemes over S, with X unramified and separated over S and Y connected. Let $f, g\colon Y \to X$ be two S-morphisms, and let $y \in Y$. Suppose that $f(y) = g(y) = x$ and that the homomorphisms of residue fields $\kappa(x) \to \kappa(y)$ induced by f and g are identical ("f and g coincide geometrically at y"). Then f and g are identical.*

This follows from Corollary I.5.3 by reducing to the case $Y = S$, replacing X by $X \times_S Y$.

Here is a particularly important variant of Corollary I.5.3.

Theorem I.5.5. *Let S be a prescheme, let X and Y be two S-preschemes, and let S_0 be a closed subprescheme of S having the same underlying space as S. Put*

$$X_0 = X \times_S S_0 \qquad \textit{and} \qquad Y_0 = Y \times_S S_0,$$

the "restrictions" of X and Y to S_0. Suppose that X is étale over S. Then the natural map

$$\mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_{S_0}(Y_0, X_0)$$

is bijective.

We are again reduced to the case $Y = S$; the assertion then follows from the "topological" description of the sections of X/Y given in Corollary I.5.3.

[9] The French TeX and the original printing give only "surjective and radicial" and cite Theorem I.5.1, which also requires étaleness. Without it the claim is false: for example, $\mathrm{Spec}(k) \to \mathrm{Spec}(k[\varepsilon]/(\varepsilon^2))$ is surjective on points, radicial, unramified, separated, and of finite type, but is not an isomorphism. The logically correct criterion is used here with the source defect disclosed.

Scholium. This result includes both a *uniqueness* assertion and an *existence* assertion for *morphisms*. It may also be expressed, when X and Y are both taken to be étale over S, by saying that the functor

$$X \longmapsto X_0$$

from the category of étale S-schemes to the category of étale S_0-schemes is *fully faithful*; that is, it establishes an *equivalence* of the first category with a *full subcategory* of the second. We shall see below that it is actually an equivalence between the first and the second (which will be an *existence theorem for étale S-schemes*). French source p. 8

The following form of Theorem I.5.5, apparently more general, is often convenient.

Corollary I.5.6 ("Extension theorem for liftings"). *Consider a commutative diagram of morphisms*

$$\begin{array}{ccc} X & \longleftarrow & Y_0 \\ \downarrow & & \downarrow \\ S & \longleftarrow & Y \end{array}$$

in which $X \to S$ is étale and $Y_0 \to Y$ is a bijective closed immersion. Then there is a unique morphism $Y \to X$ that makes the two corresponding triangles commute.

Indeed, replacing S by Y and X by $X \times_S Y$, we are reduced to the case $Y = S$; this is then the special case of Theorem I.5.5 with $Y = S$.

We should also note the following immediate consequence of Theorem I.5.1 (which we did not state as Corollary 1, so as not to interrupt the line of ideas developed after Theorem I.5.1):

Proposition I.5.7. *Let X and X' be two preschemes of finite type and flat over Y, and let $g\colon X \to X'$ be a Y-morphism. For g to be an open immersion (respectively, an isomorphism), it is necessary and sufficient that, for every $y \in Y$, the induced morphism on the fibers*

$$g \otimes_Y \kappa(y)\colon X \otimes_Y \kappa(y) \longrightarrow X' \otimes_Y \kappa(y)$$

be so.

It is enough to prove sufficiency. Since the assertion holds for surjectivity, we are reduced to the case of an open immersion. By Theorem I.5.1, one must verify that g is *radicial*, which is immediate, and that it is *étale*, which follows from Corollary I.5.9 below.

Corollary I.5.8. *(This should be moved to no. I.3.) Let X and X' be two preschemes over Y, let $g\colon X \to X'$ be a Y-morphism, let $x \in X$, and let y be its image in Y. For g to be quasi-finite (respectively, unramified) at x, it is necessary and sufficient that the same be true of $g \otimes_Y \kappa(y)$.*

Indeed, the two algebras over $\kappa(g(x))$ that one must examine to verify that the morphism is quasi-finite (respectively, unramified) at x are the same for g and for $g \otimes_Y \kappa(y)$. French source p. 9

Corollary I.5.9. *With the notation of Corollary I.5.8, suppose that X and X' are flat and of finite type over Y. For g to be flat (respectively, étale) at x, it is necessary and sufficient that $g \otimes_Y \kappa(y)$ be so.*

For "flat," the statement is included only as a reminder: it is one of the fundamental criteria for flatness.[10] For "étale," the assertion follows from this, taking Corollary I.5.8 into account.

[10]Cf. Exposé IV, no. 5.9.

I.6. Application to étale extensions of complete local rings

This number is a special case of results on formal preschemes that will have to appear in the *multiplodoque*. Nevertheless, here one can get by more cheaply, that is, without the explicit local determination of étale morphisms in no. I.7 (using the Main Theorem). This is perhaps a sufficient reason to keep the present number (even in the *multiplodoque*) in this place.

THEOREM I.6.1. *Let A be a complete local ring (noetherian, of course), with residue field k. For every A-algebra B, put*

$$R(B) = B \otimes_A k,$$

regarded as a k-algebra; it thus depends functorially on B. Then R defines an equivalence *between the category of A-algebras* finite and étale over A *and the category of algebras* of finite rank and separable *over k.*

First, the functor in question is fully faithful, as follows from the more general fact:

COROLLARY I.6.2. *Let B and B' be two A-algebras finite over A. If B is étale over A, then the canonical map*

$$\mathrm{Hom}_{A\text{-}alg}(B, B') \longrightarrow \mathrm{Hom}_{k\text{-}alg}(R(B), R(B'))$$

is bijective.

We are reduced to the case in which A is artinian (by replacing A with $A/\mathfrak{m}^n$); this is then a special case of Theorem I.5.5.

It remains to prove that for every finite separable k-algebra L (why not say "étale"? It is shorter), there is an algebra B, étale over A, such that $R(B)$ is isomorphic to L. We may suppose that L is a separable extension of k; as such, it has a generator x, that is, it is isomorphic to an algebra $k[t]/Fk[t]$, where $F \in k[t]$ is a monic polynomial. Lift F to a monic polynomial $F_1 \in A[t]$, and take

$$B = A[t]/F_1A[t].$$

I.7. Local construction of unramified and étale morphisms

PROPOSITION I.7.1. *Let A be a noetherian ring, B a finite A-algebra, u a generator of B over A, and $F \in A[t]$ such that $F(u) = 0$ (we do not assume that F is monic). Put $u' = F'(u)$, where F' is the derivative of F. Let $\mathfrak{q}$ be a prime ideal of B that does not contain u', and let $\mathfrak{p}$ be its contraction to A. Then $B_{\mathfrak{q}}$ is unramified over $A_{\mathfrak{p}}$.* French source p. 10

In other words, put $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, and $X_{u'} = \mathrm{Spec}(B_{u'})$. Then $X_{u'}$ is unramified over Y. The assertion follows from the following, more precise one.

COROLLARY I.7.2. *The different ideal of B/A contains $u'B$, and is equal to it if the natural homomorphism*

$$A[t]/FA[t] \longrightarrow B,$$

which sends t to u, is an isomorphism.

Let J be the kernel of the homomorphism $C = A[t] \to B$. This kernel contains $FA[t]$, and is equal to it in the second case considered in Corollary I.7.2. Since the homomorphism is surjective, $\Omega^1_{B/A}$ is identified with the quotient of $\Omega^1_{C/A}$ by the submodule generated by $J\Omega^1_{C/A}$ and $d(J)$ (the definition of the homomorphism d, and the computation of Ω^1 for a polynomial algebra, should have been made explicit in no. I.1). Identifying $\Omega^1_{C/A}$ with C by means of the basis dt, one obtains

$$B/(B \cdot J'),$$

where J' is the set of images in B of the derivatives of the polynomials $G \in J$; thus the different ideal is generated by J' (and it is enough to take polynomials G that generate J). Since $F \in J$, respectively since F is a generator of J, the proof is complete. (N.B. Corollary I.7.2 should be a proposition and Proposition I.7.1 a corollary.) We obtain:

COROLLARY I.7.3. *Under the hypotheses of Proposition I.7.1, suppose that F is monic and that $A[t]/FA[t] \to B$ is an isomorphism. Then $B_{\mathfrak{q}}$ is étale over $A_{\mathfrak{p}}$ if and only if $\mathfrak{q}$ does not contain u'.*

Indeed, since B is flat over A, being étale is equivalent to being unramified, and Corollary I.7.2 applies.

COROLLARY I.7.4. *Under the hypotheses of Corollary I.7.3, B is étale over A if and only if u' is invertible, or, equivalently, if and only if the ideal generated by F and F' in $A[t]$ is the unit ideal.*

The last criterion follows from the first and Nakayama's lemma (in B).

A monic polynomial $F \in A[t]$ having the property stated in Corollary I.7.4 is called a *separable polynomial.* If F is not monic, one should at least require the coefficient of its leading term to be invertible; when A is a field, this recovers the usual definition. French source p. 11

COROLLARY I.7.5. *Let B be a* finite *algebra over the* local *ring A. Suppose that $K(A)$ is infinite or that B is local. Let n be the rank of $L = B \otimes_A K(A)$ over $K(A) = k$. Then B is unramified (respectively, étale) over A if and only if B is isomorphic to a quotient of (respectively, isomorphic to)*

$$A[t]/FA[t],$$

where F is a monic separable polynomial that may be assumed (respectively, is necessarily) to have degree n.

Only necessity has to be proved. Suppose that B is unramified over A, so that L is separable over k. The hypothesis then implies that L/k has a generator ξ, and hence the ξ^i, $0 \leq i < n$, form a basis of L over k. Let $u \in B$ lift ξ. By Nakayama's lemma, the u^i, $0 \leq i < n$, generate the A-module B (respectively, form a basis of it). In particular, one can find a monic polynomial $F \in A[t]$ such that $F(u) = 0$, and B is isomorphic to a quotient of (respectively, isomorphic to) $A[t]/FA[t]$. Finally, by Corollary I.7.4 applied to L/k, F and F' generate $A[t]$ modulo $\mathfrak{m}A[t]$. Hence, by Nakayama's lemma in $A[t]/FA[t]$, F and F' generate $A[t]$.

THEOREM I.7.6. *Let A be a local ring, and let $A \to \mathcal{O}$ be a local homomorphism such that $\mathcal{O}$ is isomorphic to a localization of an algebra of finite type over A. Suppose that $\mathcal{O}$ is unramified over A. Then there exist an A-algebra B integral over A, a maximal ideal $\mathfrak{n}$ of B, a generator u of B over A, and a monic polynomial $F \in A[t]$, such that*

$$F'(u) \notin \mathfrak{n} \qquad \text{and} \qquad \mathcal{O} \simeq B_{\mathfrak{n}}$$

as A-algebras. If $\mathcal{O}$ is étale over A, one may take $B = A[t]/FA[t]$.

(Of course, the conditions displayed here are also sufficient...)

Let us first point out the pleasant corollaries.

COROLLARY I.7.7. *The algebra $\mathcal{O}$ is unramified over A if and only if it is isomorphic to a quotient of an algebra of the same form that is étale over A.*

Indeed, take $\mathcal{O}' = B'_{\mathfrak{n}'}$, where $B' = A[t]/FA[t]$ and $\mathfrak{n}'$ is the inverse image of $\mathfrak{n}$ in B'. French source p. 12

COROLLARY I.7.8. *Let $f\colon X \to Y$ be a morphism of finite type and let $x \in X$. Then f is unramified at x if and only if there is an open neighborhood U of x such that $f|_U$ factors as*

$$U \longrightarrow X' \longrightarrow Y,$$

where the first arrow is a closed immersion and the second is an étale morphism.

This is a straightforward translation of Corollary I.7.7.

Let us show how the statement of Theorem I.7.6 follows from the main assertion. By Corollary I.7.7, there is a surjective homomorphism $\mathcal{O}' \to \mathcal{O}$, where $\mathcal{O}'$ has the desired form.[11] If $\mathcal{O}$ is étale over A, then both $\mathcal{O}'$ and $\mathcal{O}$ are étale over A; the morphism $\mathcal{O}' \to \mathcal{O}$ is therefore étale by Corollary I.4.8, and hence is an isomorphism.

Proof of Theorem I.7.6. This repeats a proof from the Chevalley seminar. By the *Main Theorem*, one has $\mathcal{O} = B_{\mathfrak{n}}$, where B is a finite algebra over A and $\mathfrak{n}$ is a maximal ideal of B. Then $B/\mathfrak{n} = K(\mathcal{O})$ is a separable, hence monogenic, extension of k. If $\mathfrak{n}_i$, $1 \leq i \leq r$, are the maximal ideals of B distinct from $\mathfrak{n}$, there is therefore an element $u \in B$ that belongs to every $\mathfrak{n}_i$ and whose image in $B/\mathfrak{n}$ generates that algebra. Moreover,

$$B/\mathfrak{n} = B_{\mathfrak{n}}/\mathfrak{n}B_{\mathfrak{n}} = B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}},$$

where $\mathfrak{m}$ is the maximal ideal of A. Let us admit for the moment the following lemma.

Lemma I.7.9. *Let A be a local ring, B a finite A-algebra, and $\mathfrak{n}$ a maximal ideal of B. Let $u \in B$ be an element whose image in $B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}}$ generates that algebra over $k = A/\mathfrak{m}$, and which belongs to every maximal ideal of B distinct from $\mathfrak{n}$. Put*

$$B' = A[u], \qquad \mathfrak{n}' = \mathfrak{n} \cap B'.$$

Then the canonical homomorphism

$$B'_{\mathfrak{n}'} \longrightarrow B_{\mathfrak{n}}$$

is an isomorphism.[12]

Lemma I.7.10. *(This should have appeared as a corollary to Proposition I.7.1, before Corollary I.7.5, which it implies.) Let B be a finite A-algebra generated by an element u, and let $\mathfrak{n}$ be a maximal ideal of B such that $B_{\mathfrak{n}}$ is unramified over A. Then there is a monic polynomial $F \in A[t]$ such that*

$$F(u) = 0 \qquad \textit{and} \qquad F'(u) \notin \mathfrak{n}.$$

Indeed, let n be the rank of the k-algebra $L = B \otimes_A k$. By Nakayama's lemma there is a monic polynomial $F \in A[t]$ of degree n such that $F(u) = 0$. Let f be the reduction of F modulo $\mathfrak{m}$. Then L is isomorphic, as a k-algebra, to $k[t]/fk[t]$. Hence, by Corollary I.7.3, $f'(\xi)$ does not belong to the maximal ideal of L corresponding to $\mathfrak{n}$, where ξ denotes the image of t in L, that is, the image of u in L. Since $f'(\xi)$ is the image of $F'(u)$, the proof is complete.

Theorem I.7.6 now follows by combining Lemmas I.7.9 and I.7.10. It remains to prove Lemma I.7.9. Put $S' = B' \setminus \mathfrak{n}'$, so that $B'S'^{-1} = B'_{\mathfrak{n}'}$. We have the diagram French source p. 13

$$\begin{array}{ccccc} B & \longrightarrow & BS'^{-1} & \longrightarrow & BS^{-1} = B_{\mathfrak{n}} \\ \uparrow & & \uparrow & & \\ B' & \longrightarrow & B'S'^{-1} = B'_{\mathfrak{n}'} & & \end{array}$$

where, similarly, $S = B \setminus \mathfrak{n}$, so that $BS^{-1} = B_{\mathfrak{n}}$. There is therefore a natural homomorphism

$$BS'^{-1} \longrightarrow BS^{-1} = B_{\mathfrak{n}}.$$

[11]The French TeX and the original printing say literally "the jargon of Theorem I.7.6," write the unprimed $\mathcal{O}$ after "where," and move directly to "since $\mathcal{O}'$ and $\mathcal{O}$ are étale." Corollary I.7.7 constructs the primed algebra with the desired étale presentation, while the latter sentence concerns the theorem's final case, in which $\mathcal{O}$ is additionally assumed étale. Both points are made explicit here.

[12]The French TeX and the original printing give $B' = B[u]$ and $\mathfrak{n}' = \mathfrak{n}B'$. Since $u \in B$, the first formula would make $B' = B$, while the proof uses B' as the finite A-subalgebra generated by u and uses $\mathfrak{n}'$ as the contraction of $\mathfrak{n}$. The type-correct formulas required by the proof are used here with the source reading disclosed.

We prove that it is an isomorphism. Equivalently, the elements of S must be invertible in BS'^{-1}; equivalently, every maximal ideal $\mathfrak{p}$ of the latter ring must be disjoint from S, that is, must induce $\mathfrak{n}$ on B. Since BS'^{-1} is finite over $B'S'^{-1} = B'_{\mathfrak{n}'}$, the ideal $\mathfrak{p}$ contracts to the unique maximal ideal $\mathfrak{n}'B'_{\mathfrak{n}'}$ of $B'_{\mathfrak{n}'}$,[13] and hence to the maximal ideal $\mathfrak{n}'$ of B'. Since B is finite over B', the ideal $\mathfrak{q}$ of B induced by $\mathfrak{p}$, lying over $\mathfrak{n}'$, is necessarily maximal. It does not contain u, and is therefore equal to $\mathfrak{n}$. Here we used the fact that u belongs to every maximal ideal of B distinct from $\mathfrak{n}$.

It remains to prove that $BS'^{-1} = B'S'^{-1}$. Since the former is finite over the latter, Nakayama's lemma reduces us to proving equality modulo $\mathfrak{n}'BS'^{-1}$, and it is enough, a fortiori, to prove equality modulo $\mathfrak{m}BS'^{-1}$. But

$$BS'^{-1}/\mathfrak{m}BS'^{-1} = B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}}$$

is generated over k by u (here we use the other property of u). Thus the image of B', and a fortiori of $B'S'^{-1}$, in this quotient is the whole ring, since it is a subring containing k and the image of u.

Remark *. It should be possible to state Theorem I.7.6 for an algebra $\mathcal{O}$ that is only semilocal, so as to subsume Corollary I.7.5 as well. One would assume that $\mathcal{O}/\mathfrak{m}\mathcal{O}$ is a *monogenic* k-algebra. One could then find $u \in B$ whose image in $B/\mathfrak{m}B$ is a generator and which belongs to every maximal ideal of B not arising from $\mathcal{O}$. Lemmas I.7.9 and I.7.10 should adapt without difficulty. More generally, . . .

I.8. Infinitesimal lifting of étale schemes. Application to formal schemes

Proposition I.8.1. *Let Y be a prescheme, let Y_0 be a subprescheme, let X_0 be an étale Y_0-scheme, and let x be a point of X_0. Then there exist an étale Y-scheme X, a neighborhood U_0 of x in X_0, and a Y_0-isomorphism*

$$U_0 \xrightarrow{\sim} X \times_Y Y_0.$$

Indeed, let y be the image of x in Y_0. Applying Theorem I.7.6 to the étale local homomorphism $A_0 \to B_0$ of the local rings at y and x in Y_0 and X_0, respectively, one obtains an isomorphism French source p. 14

$$B_0 = (C_0)_{\mathfrak{n}_0}, \qquad C_0 = A_0[t]/F_0A_0[t],$$

where F_0 is monic and $\mathfrak{n}_0$ is a maximal ideal of C_0 that does not contain the class of $F_0'(t)$ in C_0. Let A be the local ring at y in Y, and let $F \in A[t]$ be a monic polynomial whose image under the surjective homomorphism $A \to A_0$ is F_0 (one lifts the coefficients of F_0). Finally, put

$$C = A[t]/FA[t],$$

and let $\mathfrak{n}$ be the inverse image of $\mathfrak{n}_0$ under the natural epimorphism

$$C \longrightarrow C \otimes_A A_0 = C_0.$$

Put

$$B = C_{\mathfrak{n}}.$$

It follows immediately from the construction and Proposition I.7.1 that B is étale over A, and there is an isomorphism

$$B \otimes_A A_0 \xrightarrow{\sim} B_0.$$

[13] The French TeX and the original printing write $\mathfrak{n}'B_{\mathfrak{n}'}$ here. The ring in the preceding clause is $B'_{\mathfrak{n}'}$, so the omitted prime on B is restored explicitly.

[14] We know from EGA, Chapter I, that there exist a Y-scheme X of finite type and a point $z \in X$ above y such that $\mathcal{O}_z$ is A-isomorphic to B.[15] Since the latter is étale over $A = \mathcal{O}_y$, by taking X small enough we may assume that X is étale over Y. Put $X'_0 = X \times_Y Y_0$. The local ring at z in X'_0 is then identified with

$$\mathcal{O}_z \otimes_A A_0 = B \otimes_A A_0,$$

and is consequently isomorphic to B_0. This isomorphism is induced by an isomorphism from a neighborhood U_0 of x in X_0 onto a neighborhood of z in X'_0 (loc. cit.); by taking X small enough, we may assume that the latter neighborhood is all of X'_0. This proves the proposition.

Corollary I.8.2. *There is an analogous statement for étale* coverings*, provided the residue field $\kappa(y)$ is infinite.*

The proof is the same, with Corollary I.7.5 in place of Theorem I.7.6.

Theorem I.8.3. *The functor considered in Theorem I.5.5 is an* equivalence *of* categories.

By Theorem I.5.5, it remains to show that every étale S_0-scheme X_0 is isomorphic to an S_0-scheme $X \times_S S_0$, where X is an étale S-scheme. The underlying topological space of X must necessarily be identical with that of X_0, with X_0 moreover identified with a closed subprescheme of X. The problem is therefore equivalent to the following one: on the topological space $|X_0|$ underlying X_0, find a sheaf of algebras $\mathcal{O}_X$ over $f_0^*(\mathcal{O}_S)$ (where $f_0\colon X_0 \to S_0$ is here regarded as a continuous map of the underlying spaces) that makes $|X_0|$ into an étale S-prescheme X, together with a homomorphism of algebras

$$\mathcal{O}_X \longrightarrow \mathcal{O}_{X_0}$$

compatible with the homomorphism

$$f_0^*(\mathcal{O}_S) \longrightarrow f_0^*(\mathcal{O}_{S_0})$$

on the scalar sheaves, and inducing an isomorphism

$$\mathcal{O}_X \otimes_{f_0^*(\mathcal{O}_S)} f_0^*(\mathcal{O}_{S_0}) \xrightarrow{\sim} \mathcal{O}_{X_0}.$$

Then X will be an étale S-prescheme whose reduction is X_0, and hence will be separated over S, since X_0 is separated over S_0; thus X answers the question. Moreover, if (U_i) is an open covering of X_0, and if a solution to the problem has been found on each U_i, it follows from the uniqueness theorem I.5.5 that these solutions glue (that is, the sheaves of algebras that define them, equipped with their augmentation homomorphisms, glue). One checks at once that the ringed space over S constructed in this way is an étale S-prescheme X, equipped with an isomorphism

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$$X_0 \xrightarrow{\sim} X \times_S S_0.$$

It is therefore enough to find a solution locally, and this is provided by Proposition I.8.1.

Corollary I.8.4. *Let S be a locally noetherian formal prescheme, equipped with an ideal of definition $\mathcal{J}$,*[16] *and let*

$$S_0 = (|S|, \mathcal{O}_S/\mathcal{J})$$

[14]At this point both the French TeX and the original printing give $B \otimes_A A_0 = A_0$. Since $B = C_{\mathfrak{n}}$ and $C \otimes_A A_0 = C_0$, base change gives $(C_0)_{\mathfrak{n}_0} = B_0$. The type-correct target B_0 is used here with the shared source defect disclosed.

[15]The French TeX and the original printing have C here. The following equality with $B \otimes_A A_0$, and the appeal to the étaleness of the localized algebra, require $B = C_{\mathfrak{n}}$; B is therefore restored explicitly.

[16]The French TeX and the original printing alternate J and $\mathcal{J}$ for this ideal of definition and its powers. Since the quotient sheaves all use the same ideal sheaf, $\mathcal{J}$ is used consistently here.

be the corresponding ordinary prescheme. Then the functor

$$\mathfrak{X} \longmapsto \mathfrak{X} \times_S S_0$$

from the category of étale coverings of S to the category of étale coverings of S_0 is an equivalence of categories.

Of course, an étale covering of a *formal* prescheme S means a covering of S, that is, a formal prescheme over S defined by a coherent sheaf of algebras $\mathcal{B}$, such that $\mathcal{B}$ is *locally free* and the residual fibers

$$\mathcal{B}_s \otimes_{\mathcal{O}_s} \kappa(s)$$

of $\mathcal{B}$ are *separable* algebras over $\kappa(s)$. If S_n denotes the ordinary prescheme

$$S_n = (|S|, \mathcal{O}_S/\mathcal{J}^{n+1}),$$

then giving a coherent sheaf of algebras $\mathcal{B}$ on S is equivalent to giving a sequence of coherent sheaves of algebras $\mathcal{B}_n$ on the S_n, equipped with a transitive system of homomorphisms

$$\mathcal{B}_m \longrightarrow \mathcal{B}_n \qquad (m \geq n)$$

that define isomorphisms

$$\mathcal{B}_m \otimes_{\mathcal{O}_{S_m}} \mathcal{O}_{S_n} \xrightarrow{\sim} \mathcal{B}_n.$$

It is immediate that $\mathcal{B}$ is locally free if and only if all the $\mathcal{B}_n$ on the S_n are locally free, and that the separability condition holds if and only if it holds for $\mathcal{B}_0$, or, equivalently, for all the $\mathcal{B}_n$. Thus $\mathcal{B}$ is étale over S if and only if the $\mathcal{B}_n$ are étale over the S_n. In view of this, Corollary I.8.4 follows at once from Theorem I.8.3.

Remark *. It was not necessary in Corollary I.8.4 to restrict oneself to the case of *coverings*. This is, however, the only case used for the moment.

I.9. Permanence properties

Let $A \to B$ be a local étale homomorphism. We examine here several cases in which a given property of A implies the same property for B, or conversely. A number of such assertions already follow from the simple fact that B is *quasi-finite* and *flat* over A, and we shall confine ourselves to "recalling" a few of them: *A and B have the same Krull dimension and the same depth* (Serre's "cohomological codimension", in the terminology still in use). It follows, for example, that *A is Cohen–Macaulay if and only if B is.* Moreover, for every prime ideal $\mathfrak{q}$ of B whose contraction to A is $\mathfrak{p}$, the ring $B_{\mathfrak{q}}$ is again quasi-finite and flat over $A_{\mathfrak{p}}$, provided that B is a localization of an A-algebra of finite type. This follows from the fact that, for a morphism of finite type, the locus where it is quasi-finite, respectively flat, is open. Furthermore, *every* prime ideal $\mathfrak{p}$ of A is the contraction of a prime ideal $\mathfrak{q}$ of B, since B is *faithfully* flat over A. It follows, for example, that *$\mathfrak{p}$ and $\mathfrak{q}$ have the same height*, and also that *A has no embedded prime ideal if and only if B has none.*

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We shall therefore restrict ourselves to assertions that are more specific to étale morphisms.

Proposition I.9.1. *Let $A \to B$ be a local étale homomorphism. Then A is regular if and only if B is regular.*

Indeed, let k be the residue field of A, and let L be that of B. Since B is flat over A and $L = B \otimes_A k$, that is, $\mathfrak{n} = \mathfrak{m}B$, where $\mathfrak{m}$ and $\mathfrak{n}$ are the maximal ideals of A and B, respectively, the $\mathfrak{m}$-adic filtration on B is identical to its $\mathfrak{n}$-adic filtration, and hence

$$\mathrm{gr}^*(B) = \mathrm{gr}^*(A) \otimes_k L.$$

It follows that $\mathrm{gr}^*(B)$ is a polynomial algebra over L if and only if $\mathrm{gr}^*(A)$ is a polynomial algebra over k. This proves the proposition. (N.B. We did not use the fact that L/k is separable.)